CARD-MRI Development Institute, Inc.
A member of CARD MRI
Department of Information and Communications Technology
Bachelor of Science in Information Systems

Clinova: An OCR-Based Document Management System with Confidence-Based Error Detection and Auto-Correction for CMDI Clinic Records

A Capstone Project
Presented to the Dean and Faculty of the College of Computer Studies
In Partial Fulfillment of the Requirements for the Degree
Bachelor of Science in Information Systems

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APPROVAL SHEET

The undersigned certify that they have read this capstone project "Clinova: An OCR-Based Document Management System with Medical Terminology Recognition, Confidence-Based Error Detection, and Auto-Correction for CMDI School Clinic Records" by **[Name of Members]**, and that, in their opinion, it is satisfactory in scope and quality as a capstone project proposal for the degree of Bachelor of Science in Information Systems.

Clinova: An OCR-Based D

ocument Management System with Medical Terminology Recognition, Confidence-Based Error Detection, and Auto-Correction for CMDI School Clinic Records

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The Problem and its Background

Introduction

Effective medical information management is essential for accurate clinical decision-making, requiring proper collection, storage, and retrieval of records (İsmayılov et al., 2026). In the Philippines, school clinic health record management remains largely paper-based and inconsistent. DepEd schools still use manual records such as logbooks and printed forms despite digitalization efforts under the Universal Health Care Act (RA 11223), while CHED institutions implement non-standardized, institution-dependent systems with varying levels of digitization.

At CMDI School Clinic, records are managed manually through logbooks, folders, and Excel encoding, resulting in slow record retrieval and frequent issues such as duplicate entries, misspellings, and encoding errors. Overall, the absence of a unified digital clinic record system highlights the need for an improved document management solution. Since a significant portion of clinic records exists in paper form, a digitization approach is necessary to convert physical documents into searchable and manageable electronic records.

Optical Character Recognition (OCR) is a technology that converts printed or handwritten text into machine-readable data and is widely used in document digitization for extracting text from scanned documents and images. Although OCR systems have improved over time and can produce reliable results in standard documents (Surmieda, 2022), they still face limitations in handling recognition errors caused by poor image quality, handwriting variability, and ambiguous characters.

Existing OCR-based systems primarily focus on text extraction and storage but often lack mechanisms for evaluating recognition confidence and identifying potentially inaccurate outputs. As a result, manual verification is still required to ensure data accuracy (Bhunia et al., 2019; Surmieda, 2022). In healthcare-related environments such as school clinics, these limitations may lead to delays in record processing and reduced reliability of patient information.

To address this gap, this study proposes Clinova: An OCR-Based Document Management System with Confidence-Based Error Detection and Auto-Correction for CMDI Clinic Records. The system aims to improve the accuracy and reliability of digitized clinic records by integrating OCR processing with confidence-based error detection and automated correction of identified errors.

Project Context

The CMDI School Clinic is responsible for managing the health records of students and personnel to ensure continuity of care, support clinical decision-making, and maintain accurate health histories.

At present, the clinic operates using a fully manual, paper-based system. Consultation records, health information, physical examination results, and clinic visit logs are recorded on paper forms and stored in folders and filing cabinets. Based on interviews and observations, the clinic manages a large volume of records that increase with student enrollment and require continuous updates as students transfer, move year levels, or leave the institution.

Records are organized by year level, program, block, and alphabetical order. However, transitions between clinic personnel have led to inconsistencies in filing practices, resulting in difficulties in retrieving older or improperly stored records. According to the current nurse, recently updated records can be retrieved quickly, while older records are more difficult and time-consuming to locate due to misplaced or inconsistent filing.

Common challenges include slow search processes, misplaced documents, and incomplete or outdated records. These challenges highlight inefficiencies in document retrieval, organization, and data consistency within the clinic's record management process.

Clinic personnel emphasized the need for a centralized digital system managed by the school nurse to improve record handling, reduce reliance on paper-based documentation, and minimize manual encoding and retrieval difficulties, highlighting the need for digitization through Optical Character Recognition (OCR).

To address these challenges, the researchers propose Clinova: An OCR-Based Document Management System with Confidence-Based Error Detection and Auto-Correction for CMDI Clinic Records, which aims to digitize clinic documents using Optical Character Recognition (OCR) while improving accuracy through confidence-based error detection and auto-correction mechanisms.

Software Description

The system, Clinova, is a web-based application designed to digitize, manage, and organize clinic records for the CMDI School Clinic, improving medical record handling through enhanced accessibility, organization, and retrieval. It improves medical record handling by converting physical clinic documents into searchable and machine-readable digital records through Optical Character Recognition (OCR) technology. By replacing manual paperwork with a digital workflow, the system enhances record accessibility, organization, and retrieval within the clinic.

Clinova integrates OCR processing to extract text from scanned or image-based clinic documents such as medical forms, consultation records, and patient files. The system applies confidence-based error detection using confidence scores to identify low-confidence OCR outputs for correction or verification, followed by auto-correction to improve accuracy and reliability.

The core system capabilities of Clinova include Optical Character Recognition (OCR) processing, confidence-based error detection, and auto-correction as an integrated pipeline for improving the accuracy and reliability of extracted clinic records.

The system is composed of two functional modules: the Administrator module and the User module. The Administrator module manages user accounts, document uploads, patient record management, and system monitoring, while the User module supports OCR-based document processing, patient information handling, and record retrieval.

Supporting features include user authentication, ID scanning, smart search, activity logging, chatbot assistance, and dark mode, which enhance usability and system efficiency but are not part of the core research contribution.

System Architecture

Clinova is an OCR-Based Document Management System with Confidence-Based Error Detection and Auto-Correction for CMDI Clinic Records. The system uses React.js, CSS, and JavaScript for the front-end interface, Next.js for server-side logic and backend functionality, and PostgreSQL for database management. It is hosted on a web server to provide centralized and secure access to clinic records and document processing functionalities.

Authorized users can access the system through multiple devices such as desktops, laptops, tablets, and smartphones. The system allows users to upload scanned or image-based clinic documents, which are then processed through an OCR pipeline.

The OCR pipeline extracts text from the uploaded documents and assigns confidence scores to recognized characters and words. Based on these confidence scores, the system performs error detection by identifying potentially inaccurate outputs. Detected errors are then subjected to an auto-correction process to improve the accuracy and reliability of the extracted data before being stored in the database.

After processing, the cleaned and validated data is stored in PostgreSQL, where it becomes searchable and retrievable through the system's document management and smart search functionality.

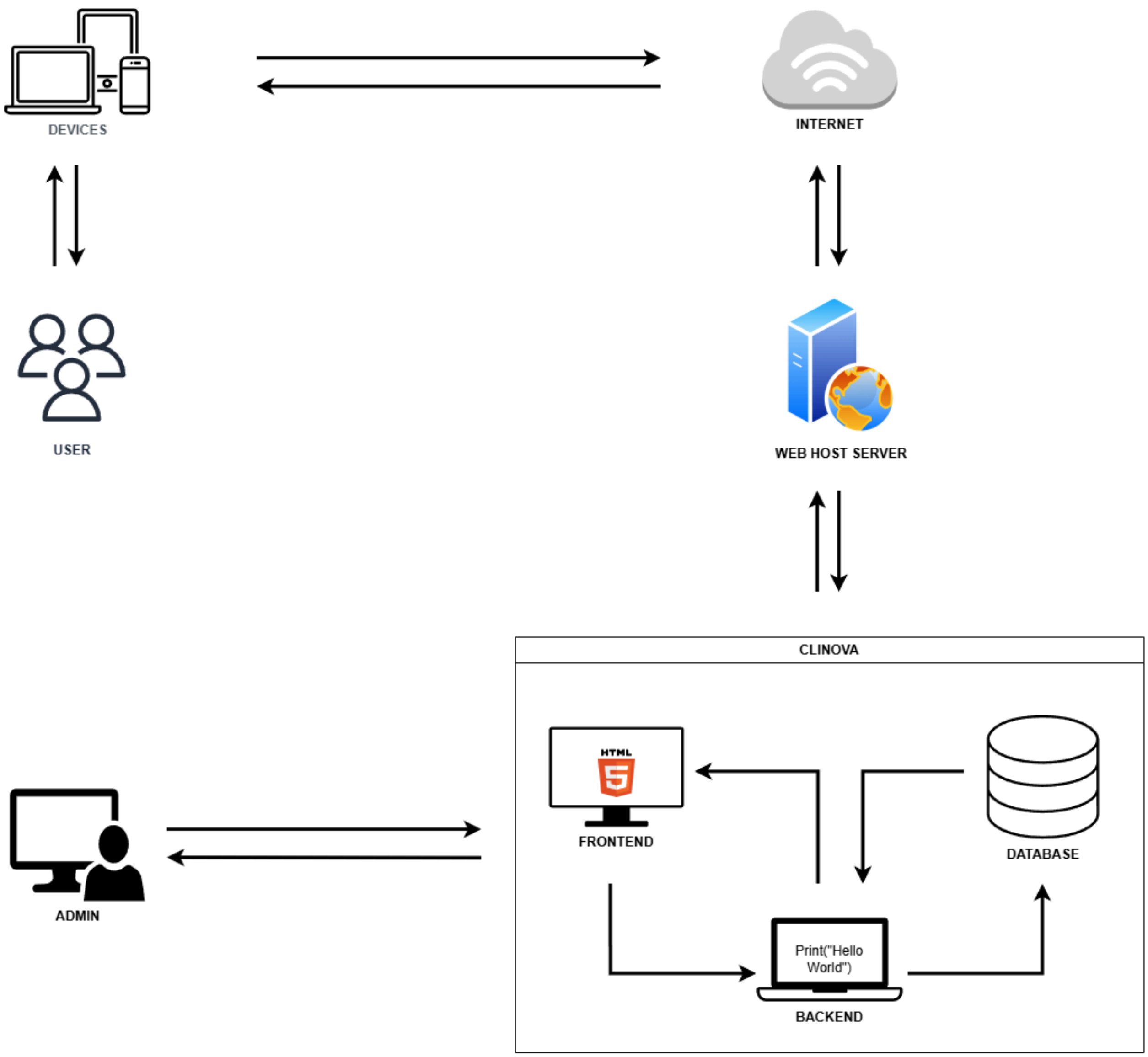

Figure1. Clinova's System Architecture

Objective of the Study

General Objectives

The main objective of the study is to develop Clinova: An OCR-Based Document Management System with Confidence-Based Error Detection and Auto-Correction for CMDI Clinic Records that enhances the digitization, organization, retrieval, and accuracy of patient records and medical documents through Optical Character Recognition (OCR), confidence-based error detection, and auto-correction mechanisms.

Specifically, the study aims to:

1. Develop an OCR-based document management system integrated with confidence-based error detection and auto-correction to improve the accuracy and reliability of digitized clinic records. The system will be capable of:
   a. Uploading, storing, retrieving, and categorizing clinic documents securely within the system.
   b. Applying confidence-based error detection to evaluate OCR outputs and identify low-confidence text for correction or verification
   c. Performing auto-correction and validation processes to improve the accuracy and reliability of extracted text
   d. Providing smart search and retrieval of clinic records
   e. Generating activity logs and document processing statistics
   f. Supporting ID scanning for patient registration and visit logging

g. Providing user authentication and access control and offering chatbot-based assistance for system navigation

2. Evaluate Clinova in terms of ISO 25010 software quality standards to assess its functionality, reliability, usability, efficiency, maintainability, and security.

3. Prepare an implementation plan for the deployment of Clinova in the CMDI School Clinic.

Conceptual Framework

The study utilizes the Input-Process-Output (IPO) Model as the conceptual framework for the development of Clinova: An OCR-Based Document Management System with Confidence-Based Error Detection and Auto-Correction for CMDI Clinic Records. The framework illustrates how the required resources (inputs) are transformed through systematic development processes to produce the desired system (output), which is evaluated using established software quality standards.

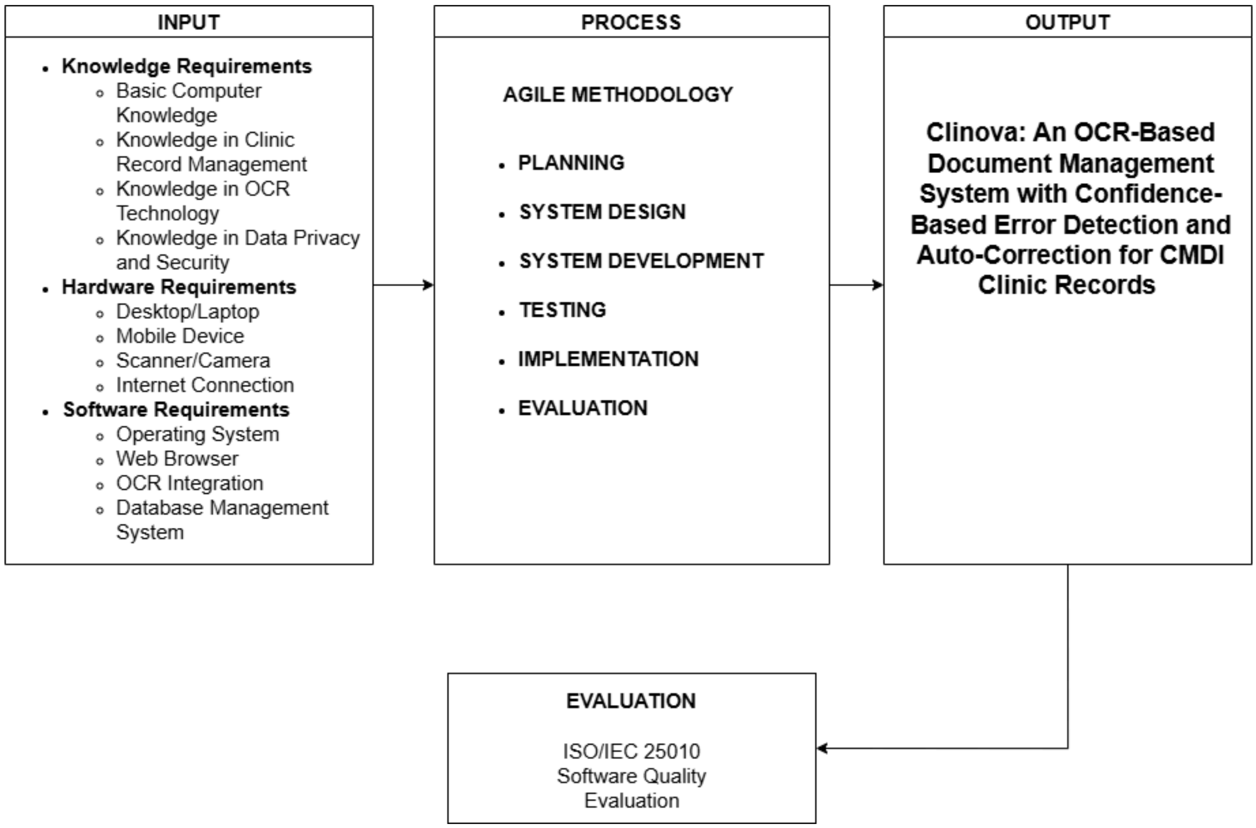

Figure 2. IPO Conceptual Framework of Clinova

Input

The input phase consists of the resources, requirements, and foundational elements necessary for the development of the system, categorized into knowledge, hardware, and software requirements.

The knowledge requirements include basic computer knowledge, understanding of clinic record management, familiarity with Optical Character Recognition (OCR) technology, and knowledge of data privacy and security. These ensure that the system aligns with healthcare documentation practices and supports secure handling of patient information.

The hardware requirements include desktop or laptop computers, mobile devices, scanners or cameras, and internet connectivity. These devices support system access, document capture, and online functionality.

The software requirements include an operating system, web browser, OCR integration tools, and a database management system. These provide the technical environment for system operation, text extraction, and secure data storage.

Process

The process phase follows the Agile Methodology, which allows iterative development and continuous improvement of the system.

The stages include:

- **Planning** – Identifying requirements of the CMDI School Clinic and defining system scope.
- **System Design** – Developing system architecture, database structure, and user interface design.
- **System Development** – Implementing core features such as document upload, OCR processing, confidence-based error detection, auto-correction, authentication, document retrieval, and search functionality.

- **Testing** – Conducting functional, integration, and usability testing to ensure system accuracy and reliability.
- **Implementation** – Deploying the system and introducing it to clinic personnel.
- **Evaluation** – Assessing system performance and user feedback based on ISO/IEC 25010 software quality standards.

Output

The output phase is the completed development of Clinova, an OCR-based document management system that digitizes clinic records, extracts text from documents, applies confidence-based error detection, and performs auto-correction to improve data accuracy.

The system provides secure storage, efficient retrieval, and organized management of clinic records. It aims to reduce manual encoding, improve document accuracy, enhance accessibility, and increase the efficiency of clinic operations.

Evaluation

Following implementation, the system will be evaluated using the ISO/IEC 25010 Software Quality Model. The evaluation will assess functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability.

The results will determine the system's effectiveness in improving clinic document management and supporting the transition to a more efficient digital records system.

Significance of the Study

The system, Clinova: An OCR-Based Document Management System with Confidence-Based Error Detection and Auto-Correction for CMDI Clinic Records, aims to improve the digitization, organization, and management of student health records at the CMDI School Clinic by providing a secure and efficient alternative to the existing paper-based filing system. By integrating Optical Character Recognition (OCR) with confidence-based error detection and auto-correction, the study contributes to improving the accuracy and reliability of digitized medical documents while reducing manual encoding and enhancing record handling efficiency.

CMDI Clinic Staff and School Nurse. The system provides faster access to organized and searchable patient records, reduces manual documentation tasks, and improves the accuracy of retrieving and managing clinic information during consultations and routine clinic operations.

CMDI Students. The system improves the efficiency of clinic services by enabling faster retrieval of health records and reducing delays in documentation processing, resulting in more organized and timely healthcare support.

CMDI Administration. The system provides organized and accessible clinic data that can support basic monitoring, reporting, and institutional decision-making related to clinic operations.

Future Researchers. The study serves as a reference for the development of OCR-based document management systems and the application of confidence-based error detection and auto-correction in similar healthcare or educational environments.

Researchers. The project provides practical experience in designing and developing an OCR-based system, applying software engineering concepts, and implementing document digitization technologies in a real-world context.

Scope and Limitation

This study focuses on the design and development of Clinova: An OCR-Based Document Management System with Confidence-Based Error Detection and Auto-Correction for CMDI Clinic Records, which aims to digitize, store, organize, and retrieve clinic records through a secure and centralized web-based platform. The system is intended for use by the Clinic Nurse and System Administrator within the CMDI School Clinic environment. It supports the conversion of paper-based clinic documents into searchable digital text through Optical Character Recognition (OCR) technology and provides a centralized database for efficient record management.

The core contribution of the system is the implementation of confidence-based error detection and auto-correction to improve the accuracy of OCR outputs. The system identifies low-confidence extracted text and applies correction mechanisms to reduce recognition errors, thereby improving the reliability of digitized clinic records. The system also includes user authentication and access control to ensure data security and proper system usage.

This study is limited to the development and evaluation of a prototype system and will assess its performance based on the ISO/IEC 25010 software quality standards. It is designed exclusively for clinic record management and is not intended for large-scale hospital or multi-department healthcare systems.

The system is limited to processing clear, standard image-based and scanned documents suitable for OCR. The accuracy of text extraction depends on the quality and clarity of the input files. While the system includes error detection and correction features, it is limited to common OCR recognition errors and may not accurately process heavily distorted, unclear, or handwritten documents.

The system does not support real-time collaboration or multi-user editing of documents. The chatbot assistance feature is limited to predefined responses for system navigation and does not provide external or professional medical advice. The analytics function is restricted to basic document processing statistics and system activity monitoring.

Definition of Terms

This section presents the important terms used in the study to help readers clearly understand their meanings within the context of the system.

Technical Definition of Terms

In technical definition, the meaning of the terms is derived from recognized dictionaries, encyclopedias, and information technology references.

Analytics – The systematic computational analysis of data or statistics used to identify patterns, interpret information, and support decision-making processes.

Clinova – The proposed Optical Character Recognition (OCR)-based document management system developed for the CMDI School Clinic that integrates confidence-based error detection and auto-correction for improving the accuracy of digitized clinic records.

Centralized System – A computer-based architecture in which data, processes, and resources are stored, managed, and controlled within a single server or central location.

Chatbot Assistance – A chatbot-based assistance feature is included as a user interface support tool that provides predefined responses to guide users in navigating the system.

Confidence-Based Error Detection – A process in OCR systems that identifies uncertain or low-confidence text recognition outputs for validation and correction.

Data Security – The protection of digital information against unauthorized access, modification, disclosure, corruption, or destruction through security measures and access control mechanisms.

Digitization – The process of converting physical or analog information into digital format for electronic storage, processing, and retrieval.

Document Management System (DMS) – A software system used to store, organize, retrieve, track, and manage digital documents and records electronically.

Health Records Management – The systematic administration, maintenance, storage, and retrieval of patient health information within a healthcare environment.

ID Scanning – A technology-driven process that captures and extracts information from identification cards using image-based scanning and recognition.

Optical Character Recognition (OCR) – A technology that converts printed or scanned text into machine-readable and editable digital text.

Patient Information – Personal and medical data associated with an individual receiving healthcare services.

Search Functionality – A software feature that enables users to efficiently locate records or documents using keywords or filters.

User Authentication – A security mechanism that verifies user identity through credentials before granting system access.

Auto-Correction – A system functionality that detects and corrects common errors in OCR-generated text to improve data accuracy and readability.

Operational Terms

The operational terms are the particular words and phrases that are used to describe the procedures or actions required to recognize and characterize the useful elements of the design project, making sure that its execution is clear.

Activity Log – A system-generated record that tracks and stores user activities within the application.

Admin Module – A system component that allows administrators to manage users, records, and system settings.

Analytics Dashboard – A graphical interface that displays summarized statistics and system performance indicators.

Database – An organized collection of data stored and managed electronically.

Dark Mode – A display setting that uses darker color themes to reduce screen brightness and improve visual comfort.

Document History – A feature that tracks changes and updates made to documents over time.

Document Upload – The process of adding digital or scanned files into the system for storage and processing.

Search Feature – A tool that allows users to retrieve records efficiently from the system.

User Interface (UI) – The visual and interactive elements through which users interact with the system.

User Module – A system component designed for clinic users to access and manage authorized functions.

User Settings – A section where users can manage account preferences and system configurations.

Chapter II

Review on Related Literatures and Studies

This chapter contains the accumulated information that the researchers gathered from different articles, books, and journals that will substantiate the study's relevance.

Related Literatures

Foreign Literature

The digitization of medical records is a critical driver for efficiency in the modern healthcare industry. Optical Character Recognition (OCR) technology enables the conversion of unstructured data, such as printed or handwritten clinic logs, into structured, machine-readable text. Modern OCR systems serve as the foundation for streamlining workflows, automating data extraction, and increasing productivity in document management. Converting physical documents into a digital format allows healthcare personnel to search and retrieve critical patient information within seconds, which significantly improves clinical operations, data archiving, and compliance [1].

However, despite achieving high accuracy rates in ideal conditions, OCR systems still encounter significant challenges when processing handwritten text or low-quality medical documents. Misclassification of characters or missing text can compromise the integrity of health records and lead to misinterpretation. To mitigate these risks, modern OCR solutions incorporate confidence scoring mechanisms. A confidence score indicates the estimated probability that the extracted text is correct. By establishing confidence thresholds, systems can automatically accept highly accurate predictions while flagging uncertain outputs for human verification, thereby ensuring accuracy and reducing data leakage for critical healthcare data [2].

To further enhance text recognition accuracy, particularly in medical contexts, the integration of post-processing techniques is highly necessary. Recent studies emphasize the importance of lexical analysis using medical dictionaries to correct misspelled drug names and common healthcare terms automatically. This contextual reasoning allows the system to resolve ambiguities and distinguish similar-looking characters based on their placement within a medical document. Applying such algorithms helps bridge the gap between simple text extraction and intelligent medical document processing [3].

Furthermore, the automatic detection and correction of medical errors are essential for validating healthcare documentation. Integrating intelligent detection frameworks ensures the medical coherence of clinical texts, which directly enhances patient care outcomes by preventing reliance on flawed OCR data [4]. Technologies like Robotic Process Automation (RPA) combined with OCR have shown high success rates in automatically flagging discrepancies and reducing the time spent on manual post-review processes by over 50% [5]. Through the implementation of similar confidence-based error detection and medical terminology auto-correction, the proposed document management system can effectively address the inefficiencies of manual record-keeping in educational clinics.

Local Literature

In the Philippines, digital transformation initiatives have encouraged the adoption of document management systems across government agencies, educational institutions, and healthcare facilities. Local studies have demonstrated the growing importance of Optical Character Recognition (OCR) technology in improving document digitization, storage, and retrieval processes. For example, the Blockchain-Based Barangay Document Management System with OCR Data Extraction developed by Abubo et al. (2024) integrated OCR technology with blockchain to improve document security, transparency, and accessibility in local government units. The study highlighted how OCR can efficiently convert paper-based documents into searchable digital records while supporting modern records management practices.

Similarly, OCR-enhanced document management solutions have been explored to improve information accessibility and retrieval efficiency. Surmieda (2022) developed an OCR-Enhanced Digital Asset Management System that enabled users to search and retrieve digitized documents more effectively. The study demonstrated that OCR technology significantly enhances document discoverability and reduces the time required to locate relevant information. Likewise, Censoro (2021) developed a Document Management System using OCR, clustering, watermarking, and QR coding algorithms to improve document organization, security, and retrieval. The integration of OCR with intelligent classification techniques allowed documents to be categorized and managed more efficiently, supporting organizational productivity.

Within the healthcare sector, the digitization of medical records has also received considerable attention. Ongkeko et al. (2016), through the implementation of the Community Health Information and Tracking System (CHITS), demonstrated the benefits of Electronic Medical Records (EMR) in Philippine healthcare facilities. The study found that digital record systems improve patient information management, facilitate faster access to medical records, and support healthcare service delivery. These findings emphasize the importance of transitioning from traditional paper-based systems to electronic records in healthcare environments.

Furthermore, educational institutions have adopted electronic document management systems to improve administrative efficiency and records preservation. Antenor Jr. and Pomperada (2025) developed the Recoletos de Bacolod Graduate School Docuware, an electronic document management system designed to modernize document archiving and retrieval processes. The study reported improvements in document accessibility, security, and storage management, demonstrating the value of digital solutions in handling institutional records.

Despite the advancements presented in these local studies, several limitations remain apparent. Most OCR-based systems focus primarily on document digitization, storage, classification, and retrieval, assuming that the extracted text is accurate. Few studies incorporate confidence scoring, OCR error detection, or automated correction mechanisms to validate extracted information. Additionally, healthcare-related systems such as CHITS emphasize electronic record management but do not specifically address the conversion of scanned medical documents through OCR or the recognition of medical terminologies. None of the reviewed local studies integrate medical terminology validation with OCR technology, leaving a gap in ensuring the accuracy and reliability of digitized healthcare records.

These limitations highlight the need for a more advanced document management solution tailored to healthcare environments. By incorporating OCR technology, confidence-based error detection, and medical terminology recognition, the proposed Clinova system seeks to address the shortcomings of existing local systems and improve the accuracy, reliability, and efficiency of healthcare document management.

Related Studies

Foreign Studies

The integration of Artificial Intelligence (AI) and Optical Character Recognition (OCR) has significantly enhanced document processing, automation, and information extraction across various domains. Serugendo et al. (2024) highlighted the use of AI-powered OCR in streamlining tax and administrative document management through an intelligent document management system. The study emphasized improvements in document organization, accessibility, and processing efficiency while reducing manual workload. Despite these advantages, its application remains limited to administrative and tax-related documents and does not extend to medical terminology recognition or school clinic record systems.

In healthcare applications, Sharma et al. (2024) examined image-based extraction of prescription information using OCR-Tesseract combined with image preprocessing techniques.

The study demonstrated high recognition accuracy and reduced manual encoding in medical documentation. However, it lacks confidence-based error detection and auto-correction mechanisms, which are essential for ensuring reliability in healthcare information systems.

Similarly, Soeno et al. (2024) developed an OCR-based system to reduce the time required for recording vital signs and prescriptions in clinical environments. The findings showed improved efficiency in data capture and documentation processes. Despite this improvement, the system does not incorporate document management capabilities or intelligent correction features for inaccurate OCR outputs.

In the field of data security, Psarra et al. (2024) explored a permissioned blockchain approach to enhance the security of electronic health records. The study highlighted improved data protection and controlled access to sensitive medical data. However, it does not incorporate OCR functionality or text correction mechanisms.

Advancements in OCR post-processing were presented by Shim, Hong, and Lim (2026) through the "Revise" framework, which applies AI-based structured correction to OCR-generated text. The results showed a significant reduction in recognition errors and improved text quality. Nonetheless, the framework is not specifically designed for clinic records or medical terminology processing.

From an evaluation perspective, Liu et al. (2024) introduced OCRBench, a benchmarking framework for assessing modern OCR systems. The study revealed persistent challenges in handling real-world and handwritten documents. Still, it focuses solely on evaluation and does not provide correction or user interaction features.

Zhang et al. (2024) proposed CLOCR-C, a language model–based OCR correction system that enhances text accuracy through contextual understanding. The study demonstrated effective error reduction; however, it operates as a backend process without user-facing validation or interaction.

Nguyen et al. (2025) investigated synthetic data generation to improve OCR error correction using deep learning approaches. The study improved correction performance in structured documents but lacks user-centered validation tools or interactive correction interfaces.

Kumar et al. (2025) examined probabilistic modeling techniques for estimating uncertainty in OCR outputs. The study concluded that confidence estimation improves OCR reliability by identifying uncertain predictions, yet it does not provide mechanisms for user-guided correction.

Lastly, Hemmer et al. (2024) developed a confidence-aware OCR error detection system based on word-level confidence scoring. The study confirmed a strong correlation between low-confidence outputs and OCR errors. However, it lacks user interface integration and real-world deployment in operational systems.

Local Studies

In the local context, OCR and document management systems have been widely adopted to improve administrative and institutional processes. Jayoma et al. (2020) developed an OCR-based document archiving and indexing system for DSWD Caraga that digitizes government records using OCR technology. The system improved storage and retrieval efficiency but assumes OCR outputs are accurate and lacks error correction mechanisms.

Surmieda (2022) proposed an OCR-enhanced digital asset management system to improve document searchability in a digital library environment. Findings showed improved retrieval speed and accessibility; however, it lacks confidence scoring and validation features to assess OCR accuracy.

Tumasis (2022) developed an OCR-based record management system for accounting firms that automates document processing and reduces manual encoding. While operational efficiency improved, the system does not address OCR inaccuracies that may affect data reliability.

Censoro (2021) introduced a document management system integrating OCR and TF-IDF-based text classification, improving document organization and retrieval efficiency. However, the study emphasizes classification rather than accuracy validation or correction.

Almacen (2021) examined electronic document management systems in Philippine healthcare institutions and found that EDMS improves accessibility, storage, and workflow efficiency. Nonetheless, it lacks intelligent OCR processing and error detection features.

In healthcare applications, Goh et al. (2025) developed a web-based clinic management system integrated with sentiment analysis to evaluate patient satisfaction. Using machine learning and natural language processing, the system improved service evaluation and decision-making. Although effective in enhancing operational performance, it does not involve OCR-based document digitization or record conversion.

Similarly, Aller et al. (2024) developed a clinic record system for Senior High School clinics that replaces manual record-keeping using Word and Excel. The system improved accessibility and organization of student health records, including medical history and consultation data. However, it does not incorporate OCR technology or intelligent error correction mechanisms.

Synthesis

Based on the reviewed literature and studies, Optical Character Recognition (OCR) technology plays a significant role in improving document digitization, storage, and retrieval across healthcare, education, and government sectors. Foreign studies emphasize that integrating artificial intelligence, confidence scoring, and OCR-based processing improves text recognition accuracy and reduces manual workload in document management systems.

However, despite these advancements, most existing systems focus primarily on text extraction and document organization, assuming that OCR outputs are accurate. Limited studies incorporate mechanisms such as confidence-based error detection, automated correction, and domain-specific validation for medical terminology.

Similarly, local studies show that OCR-based and electronic document management systems improve accessibility, retrieval efficiency, and administrative workflows. Nevertheless, these systems largely lack intelligent validation features that assess OCR accuracy and correct potential errors in extracted data, particularly in healthcare-related applications.

Therefore, a research gap exists in the development of OCR-based systems that integrate confidence-based error detection and auto-correction specifically for healthcare environments. The proposed system, Clinova, addresses this gap by combining OCR processing with intelligent validation mechanisms to improve the accuracy, reliability, and usability of digitized clinic records.

Chapter III

Research Design and Methodology

Requirement Analysis

The study adopted a requirement analysis approach to identify the functional and non-functional requirements of the proposed Clinova system. Data were gathered through interviews with clinic personnel, direct observation, and review of existing clinic record management procedures at the CMDI School Clinic.

The researchers examined the current workflow involving patient registration, record filing, document retrieval, record updating, and report generation. Findings revealed that the clinic primarily relies on paper-based records, logbooks, folders, and spreadsheet encoding, resulting in inefficiencies in record management and information retrieval.

The identified problems included:

- Time-consuming retrieval of patient records and clinic documents
- Dependence on manual logbooks, paper forms, and physical filing systems
- Inconsistent organization of records due to varying filing practices
- Misplaced or difficult-to-locate clinic documents
- Incomplete and outdated patient information
- Repetitive manual encoding and updating of student records
- Limited accessibility to archived and historical records
- Lack of a centralized digital repository for clinic documents
- Difficulty in searching and tracking patient information efficiently

- ➔ Risk of data entry errors during manual record management

- ➔ Absence of automated document digitization and validation mechanisms
- ➔ Limited monitoring and tracking of record management activities

Requirement Documentation

Functional Requirements

The functional requirements describe the specific features and services that the proposed Clinova system must provide.

The system shall:

- ➔ Allow authorized users to securely log in and access the system.
- ➔ Manage user accounts and role-based access permissions.
- ➔ Store, update, organize, and retrieve patient records electronically.
- ➔ Upload and manage scanned clinic documents and medical forms.
- ➔ Convert scanned documents into machine-readable text using Optical Character Recognition (OCR).
- ➔ Extract relevant information from uploaded clinic documents.
- ➔ Recognize and validate medical terminologies through a predefined medical terminology database.
- ➔ Detects uncertain OCR outputs using confidence-based error detection mechanisms.

- ➔ Automatically suggest corrections for incorrectly recognized words and medical terms.
- ➔ Allow users to review, verify, and edit OCR-generated results before final storage.
- ➔ Categorize documents according to predefined clinic classifications.
- ➔ Provide smart search functionality for efficient retrieval of patient records and documents.
- ➔ Support ID scanning for capturing patient information and reducing manual encoding.
- ➔ Generate clinic reports and basic statistical summaries.
- ➔ Display analytics related to patient visits, document usage, and system activities.
- ➔ Maintain activity logs for monitoring user actions and system usage.
- ➔ Provide chatbot assistance using predefined clinic information and stored system data.
- ➔ Perform backup and recovery processes for data protection and continuity.

Non-Functional Requirements

The non-functional requirements define the quality attributes and operational standards of the proposed Clinova system.

Security – The system must protect confidential patient records through secure authentication, access control, activity logging, and data protection mechanisms.

Accuracy – The system must provide reliable OCR processing, medical terminology recognition, confidence-based error detection, and auto-correction to improve the accuracy of digitized clinic records.

Reliability – The system must maintain stable operation and ensure that records remain available, consistent, and protected from loss.

Performance Efficiency – The system should provide fast processing of OCR tasks, record retrieval, document searching, and report generation.

Usability – The interface should be user-friendly, intuitive, and easy for clinic personnel to learn and operate.

Accessibility – The system should be accessible through standard web browsers and supported devices used within the CMDI School Clinic.

Maintainability – The system should support future updates, debugging, and enhancements with minimal disruption to operations.

Data Integrity – The system must maintain accurate, complete, and consistent patient records while preventing duplication and unauthorized modification.

Availability – The system should ensure that authorized users can access clinic records whenever needed for daily clinic operations.

These requirements ensure that Clinova provides a secure, accurate, reliable, and efficient platform for digitizing, organizing, retrieving, and managing clinic records within the CMDI School Clinic.

Design of Software

The Use Case Diagram illustrates the interactions between Medical Staff, Administrators, and the Clinova system. Medical Staff can upload, search, and retrieve medical documents, while Administrators manage users, generate reports, view analytics, monitor activity logs, and perform database backups. The Upload Medical Document use case includes OCR processing functions such as text extraction, medical terminology recognition, OCR error detection, auto-correction, and document storage.

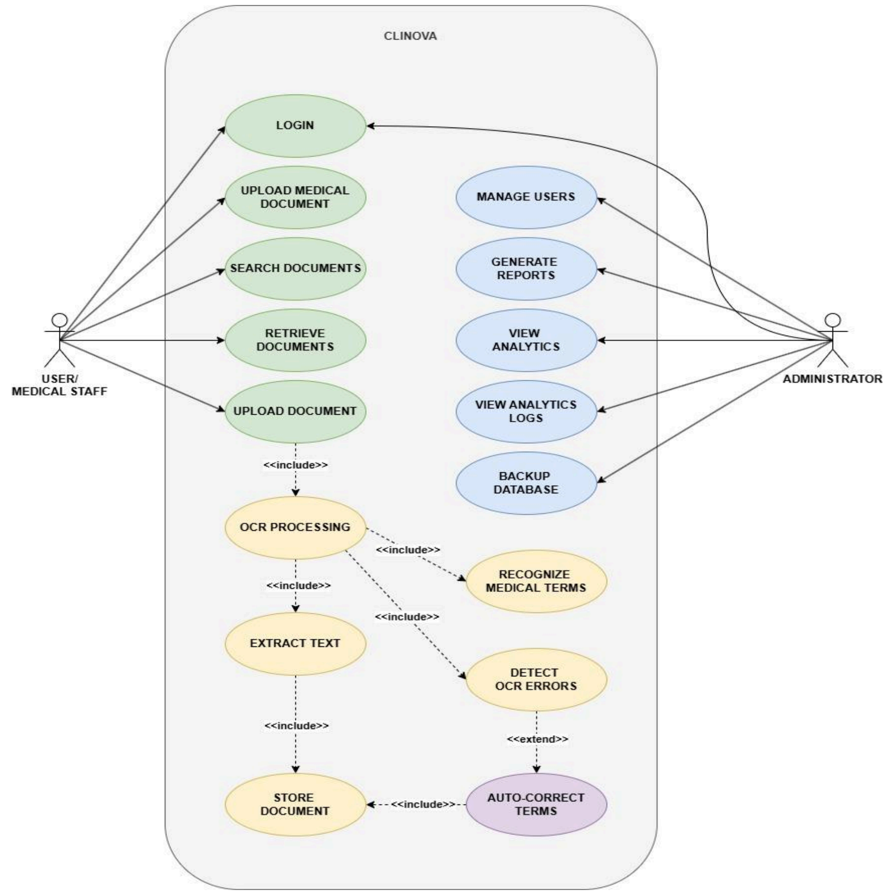

Figure 3.1 Use Case Diagram

The system architecture of Clinova illustrates the overall structure and interaction of its core components in processing, managing, and retrieving clinic documents. The system follows a client-server architecture where users (clinic staff and administrators) access the system through a web-based interface.

The frontend serves as the user interface for uploading documents, scanning patient IDs, searching records, and viewing results. All user requests are processed by the backend system, which contains the main functional modules of Clinova.

These modules include authentication and access control, patient and document management, Optical Character Recognition (OCR) processing, medical terminology recognition, confidence-based error detection, and auto-correction. After processing, data is stored in the centralized database for secure storage and efficient retrieval.

The system also includes additional services such as search and retrieval, activity logging, analytics and reporting, and chatbot assistance to support users in managing clinic records effectively.

Overall, the architecture ensures secure, efficient, and accurate processing of medical documents while improving the speed and reliability of clinic operations through automation and intelligent data processing.

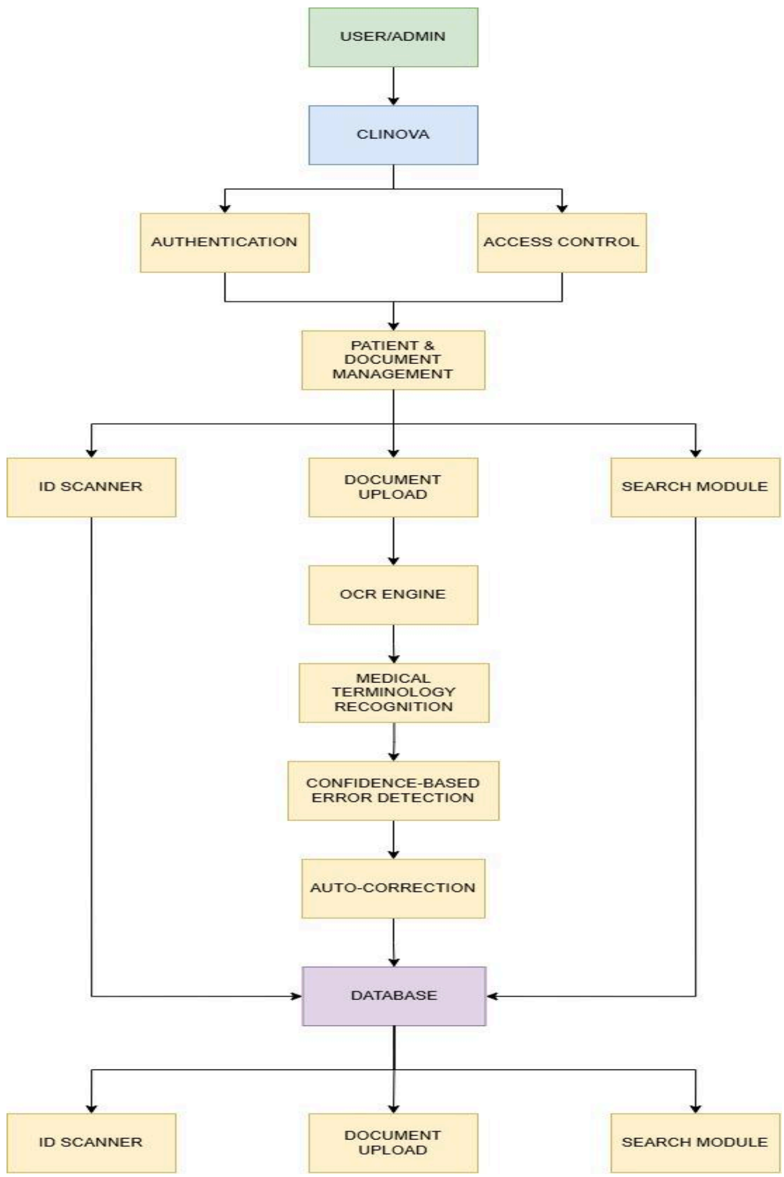
USER/ADMIN
CLINOVA
AUTHENTICATION
ACCESS CONTROL
PATIENT & DOCUMENT MANAGEMENT
ID SCANNER
DOCUMENT UPLOAD
SEARCH MODULE
OCR ENGINE
MEDICAL TERMINOLOGY RECOGNITION
CONFIDENCE-BASED ERROR DETECTION
AUTO-CORRECTION
DATABASE
ID SCANNER
DOCUMENT UPLOAD
SEARCH MODULE

Figure 3.2 System Architecture Diagram

The context diagram presents Clinova as a single system process that interacts with external entities, specifically the User (clinic staff) and the Admin. It shows the system boundary and how data flows between the users and the system.

The User provides inputs such as login credentials, document uploads, patient information through ID scanning, and search requests. In return, the system processes these inputs using its internal mechanisms and provides outputs such as OCR-extracted text, patient records, and search results.

The Admin interacts with the system by managing user accounts, monitoring system activities, and requesting reports. The system responds by generating analytics, activity logs, and administrative reports.

This diagram focuses only on external interactions with Clinova and does not include internal processes such as OCR processing or database operations, which are detailed in the Level 1 Data Flow Diagram.

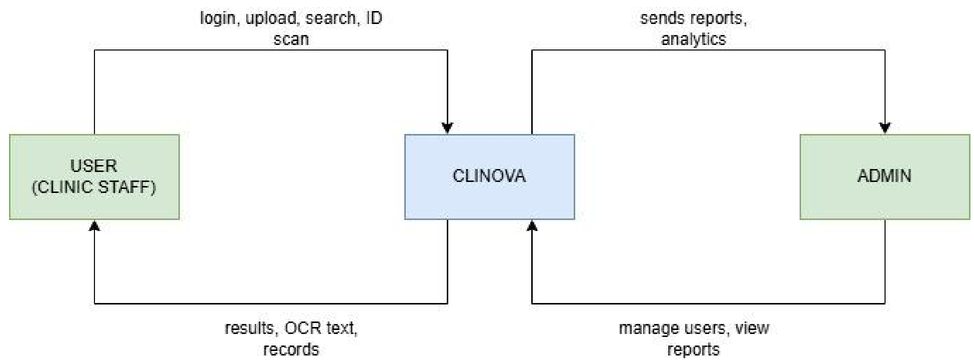

Figure 3.3 Data Flow Diagram Level 0 (Context Diagram)

The Data Flow Diagram Level 1 shows how clinic staff interact with the Authentication and Document & Patient Management processes to upload, scan, search, and manage medical documents. The system processes scanned documents through the OCR Engine, Medical Term Recognition, Error Detection, and Auto-Correction modules before storing the extracted information in the database. The database serves as the central repository of medical records and supports the Search, Analytics, and Activity Logs processes. Administrators access reports and logs generated by the system to monitor activities and manage system operations.

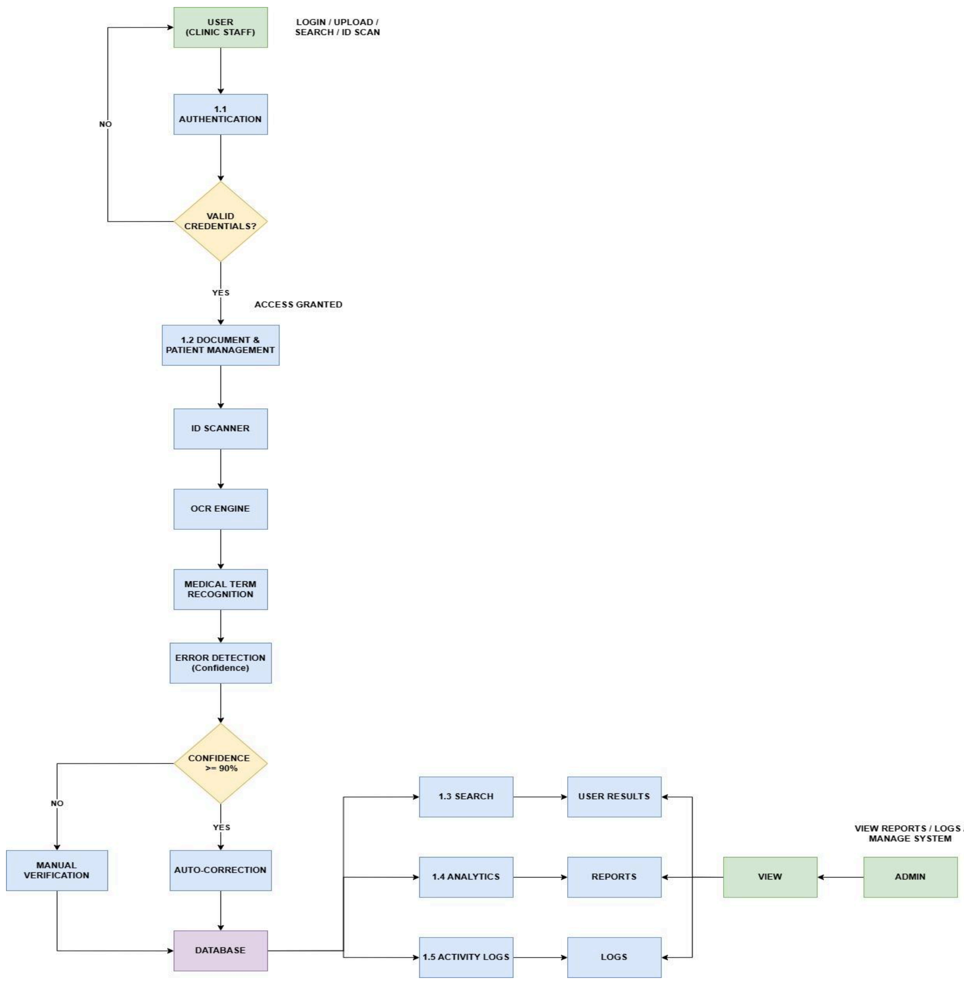

Figure 3.4 Data Flow Diagram Level 1

The Entity Relationship Diagram (ERD) illustrates the database structure of Clinova and the relationships among its major entities. The system consists of six primary entities: Users, Patients, Medical Documents, OCR Results, Medical Terms, and Activity Logs.

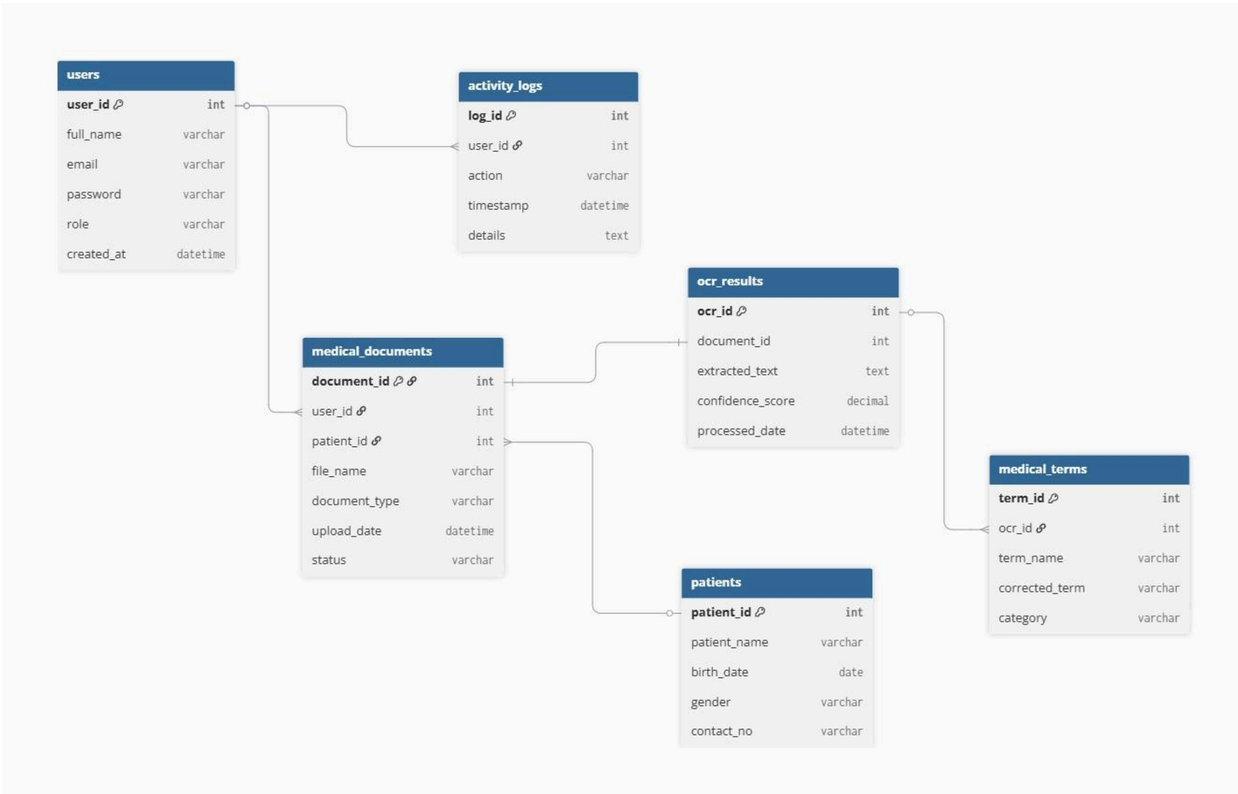

Figure 3.5 Entity Relationship Diagram (ERD)

The Users entity stores information about medical staff and administrators who access the system. Each user can upload multiple medical documents and generate multiple activity log records. The Activity Logs entity records user actions such as login attempts, document uploads, document retrieval, and administrative activities for monitoring and auditing purposes.

The Patients entity contains patient information associated with medical records. A patient may have multiple medical documents stored within the system. The Medical Documents entity serves as the central repository for uploaded medical records and contains information such as file name, document type, upload date, and processing status.

After a document is uploaded, it undergoes Optical Character Recognition (OCR) processing. The extracted text and processing details are stored in the OCR Results entity. Each medical document produces a corresponding OCR result containing the recognized text and confidence score generated by the OCR engine.

The Medical Terms entity stores medical terminologies identified from the OCR-extracted text. This entity supports Clinova's medical terminology recognition feature by recording detected terms, corrected terms, and their respective categories. One OCR result may contain multiple recognized medical terms.

Overall, the ERD demonstrates how Clinova manages user activities, patient records, document storage, OCR processing, and medical terminology recognition through a structured relational database design. This design ensures efficient document retrieval, data integrity, and accurate management of digitized medical records.

The User Flowchart illustrates how new users create an account by providing the required information, which is validated and checked for duplicate records before being stored in the database. After successful registration, users are directed to the login page. The User Login Flowchart describes the authentication process wherein user credentials are verified against the database before access is granted. These processes ensure that only registered and authorized users can access Clinova.

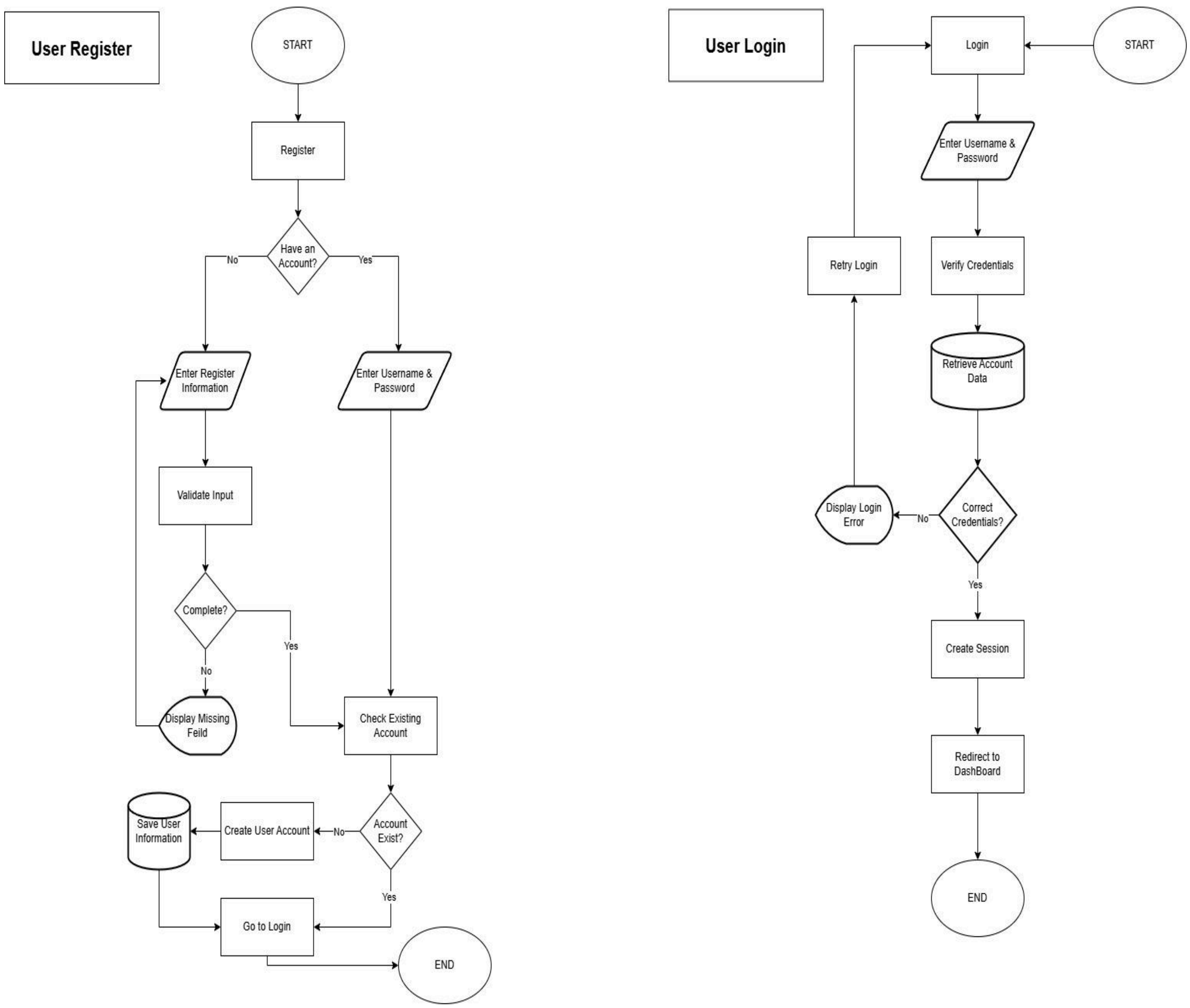

Figure 3.6 Flowchart (User)

The User Management: User Flowchart illustrates how users navigate various system modules, including Add Patient, Documents, Upload Documents, Analytics, History, and Settings. The Add Patient Flowchart details the process of creating a patient record by validating submitted information and checking for existing records before saving data to the database. Together, these processes support efficient patient information management and system navigation.

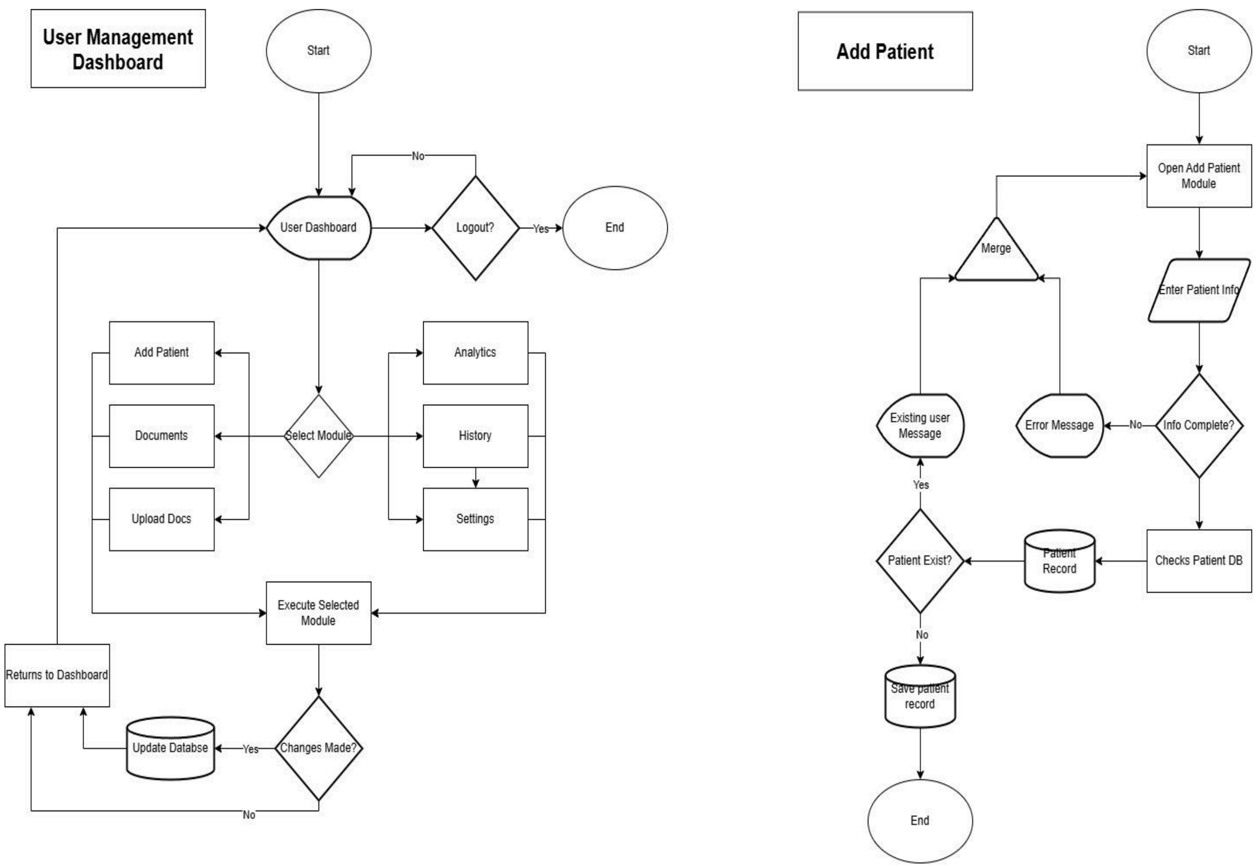

Figure 3.7 Flowchart (User management: User)

The Admin Flowchart shows how administrator credentials are validated and verified against records stored in the database. The system checks account existence and password accuracy before granting access. Upon successful authentication, a session is created, login activity is recorded, and the administrator is redirected to the dashboard. This process ensures secure access to administrative functions within the system.

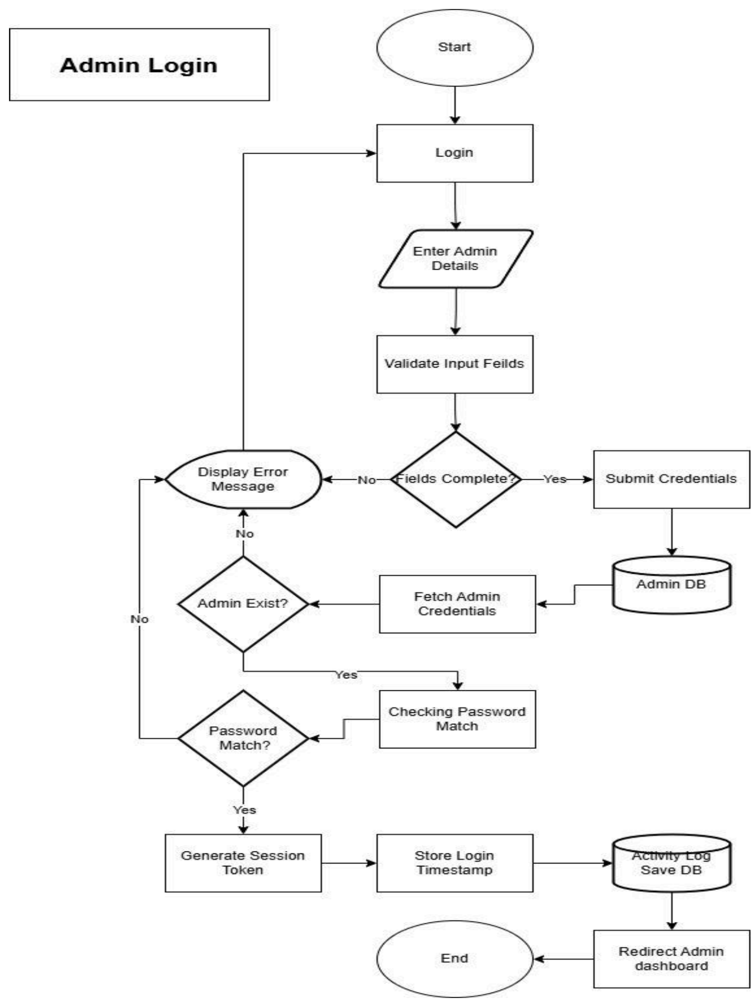

Figure 3.8 Flowchart (Admin)

The User Management: Admin Flowchart illustrates how administrators access and manage modules such as Manage Users, Manage Patients, Manage Documents, Analytics, History Logs, and System Settings. The Admin Manage User Flowchart describes the procedures for adding, editing, deactivating, and viewing user accounts while ensuring that all changes are validated and recorded in the system. These processes enable effective administration, user management, and monitoring of system activities.

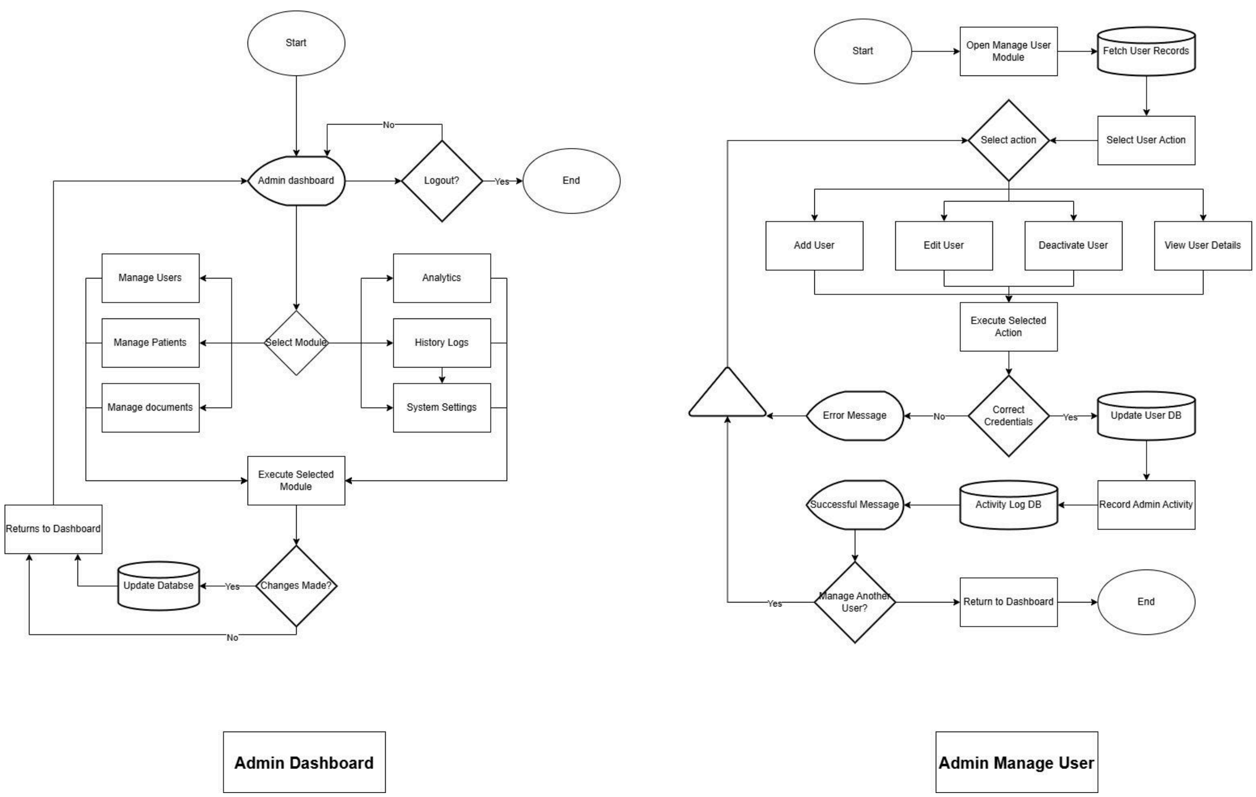

Figure 3.9 Flowchart (User management: Admin)

Development and Testing

The implementation of Clinova follows an Iterative Software Development Life Cycle (SDLC) model, which emphasizes continuous prototyping, software evaluation, and refinement. Utilizing an iterative approach allows the researchers to deploy a working version of the platform early in the development cycle and steadily optimize the underlying OCR confidence scoring algorithms. The initial requirements analysis involved mapping the manual logbook procedures and common paper forms used at the CMDI School Clinic to design the baseline database schemas. Subsequent coding and integration phases focused on constructing the Next.js server actions, deploying the database tables, and integrating the physical ID scanning functionalities to ensure seamless hardware-to-software data capture.

The verification and evaluation phases are divided into Alpha testing, Beta testing, and a standardized final evaluation. Alpha testing is conducted in a controlled environment by the developers to identify syntax bugs, database connection drops, and threshold calibration issues in the medical auto-correction engine. Following technical optimization, Beta testing will be deployed directly at the CMDI School Clinic, allowing the primary clinic nurse and staff to interact with the platform during daily student consultations. To systematically measure the system's performance, an evaluation instrument based on the ISO/IEC 25010 software quality model will be administered to both IT professionals and end-users. This framework will formally grade the Clinova prototype across critical quality attributes, including functional suitability, performance efficiency, usability, and data security, to guarantee full compliance with institutional standards.

Description of the Proptotype

The prototype of Clinova is developed as a web-based document management system engineered specifically to modernize the health record workflows of the CMDI School Clinic. The user interface is built using a component-based architecture which provides a highly responsive environment that allows clinic staff to toggle features like Dark Mode smoothly across multiple devices. The back-end infrastructure handles data pre-processing, text extraction pipelines, and security protocols, ensuring that the system can process large-scale files without sacrificing performance. To ensure data durability and support complex relational logic; such as linking student profiles to their historical medical records and activity logs, the prototype integrates a relational database management system capable of handling structured health queries efficiently.

The functional modules of the prototype are divided into the Admin Module and the User Module, which directly host the core OCR and Medical Auto-Correction Engine. When a scanned medical document or health form is uploaded, the OCR pipeline isolates textual characters and assigns a real-time confidence score to calculate the accuracy of the conversion. Text segments that fall below the acceptable confidence threshold are run through a local validation string linked directly to a predefined medical dictionary stored within the database. If a common transcription error is recognized in a specialized term or drug name, the system triggers an auto-correction or flags the word for manual validation by the nurse. This precise pipeline reduces manual verification workloads while preserving the semantic accuracy of critical clinical entries.

Implementation Plan Deployment

The implementation of Clinova: An OCR-Based Document Management System with Medical Terminology Recognition, Confidence-Based Error Detection, and Auto-Correction for CMDI School Clinic Records will follow a structured and phased approach to ensure a smooth transition from development to actual utilization within the CMDI School Clinic.

The implementation plan is divided into four major phases: system preparation, pilot deployment, full deployment, and post-deployment support. During the system preparation phase, the researchers will finalize the system setup, including database configuration, Optical Character Recognition (OCR) model integration, user account setup, and security access controls. This phase ensures that the system is fully functional and stable prior to deployment.

In the pilot deployment phase, the system will be introduced to the school clinic nurse as the primary end user of the system. This phase aims to evaluate the system’s usability, OCR accuracy, confidence-based error detection, and auto-correction functionality in an actual clinic workflow environment. Feedback gathered from the clinic nurse will be used to identify system issues, refine features, and improve overall performance.

Following successful pilot testing, full deployment will be implemented within the CMDI School Clinic. At this stage, Clinova will be utilized as the primary platform for managing clinic records, including student health information, consultation logs, and medical documents. The system will serve as a centralized repository that replaces or enhances the existing manual and semi-digital record-keeping processes.

To ensure proper system utilization, a brief orientation and training session will be conducted for the clinic nurse. This will include system navigation, document uploading procedures, OCR processing workflow, error validation techniques, and data retrieval functions. User manuals and basic troubleshooting guidelines will also be provided to support system usage.

After deployment, the system will undergo continuous monitoring and maintenance to ensure stability, security, and performance optimization. Updates will be applied as necessary, particularly to improve OCR accuracy, medical terminology recognition, and overall system efficiency based on user feedback.

Implementation Results

The implementation of Clinova is expected to produce significant improvements in the management of clinic records at CMDI School Clinic. Based on system design objectives and pilot testing expectations, the following results are anticipated:

First, the system is expected to significantly reduce the time required for retrieving and encoding patient records. Through OCR technology, printed and handwritten clinic documents will be converted into digital text, allowing faster search and access compared to the manual logbook system.

Second, the integration of confidence-based error detection is expected to improve the reliability and accuracy of extracted data. Low-confidence OCR outputs will be automatically flagged for verification, reducing the likelihood of incorrect medical entries in the system.

Third, the auto-correction and medical terminology recognition features are expected to enhance the consistency of clinical records by minimizing spelling errors and misinterpretation of medical terms commonly found in handwritten documents.

Fourth, the centralized database structure is expected to improve data organization and reduce issues related to duplication, misplacement, and incomplete records, which are common in manual filing systems.

Lastly, overall clinic workflow efficiency is expected to improve, allowing clinic personnel to focus more on patient care rather than administrative tasks. The system is also expected to enhance data accessibility, security, and long-term record management within the CMDI School Clinic environment.